Design and Create a Normalized Database Schema (3NF)

ER Relationships

```
Users (1) --------< (M) Posts
  |                    |
  v                    v
(M) Comments >-------- (1) Posts

Posts (M) ------< Post_Tags >------ (M) Tags
```

Relationships: One User can create many Posts; One User can write many Comments; One Post can have many Comments; Posts and Tags have a many-to-many relationship through Post_Tags.

SQL DDL

```
CREATE TABLE users (...);
CREATE TABLE posts (... FOREIGN KEY (user_id) REFERENCES users(user_id));
CREATE TABLE comments (... FOREIGN KEY (post_id) REFERENCES posts(post_id), FOREIGN KEY (user_id) RE
CREATE TABLE tags (...);
CREATE TABLE post_tags (post_id INT NOT NULL, tag_id INT NOT NULL, PRIMARY KEY(post_id, tag_id));
```

Index Recommendation

```
CREATE INDEX idx_comments_post_id ON comments(post_id);
```

Justification: Speeds retrieval of comments for a specific post and improves join performance.

3NF Justification

1NF: Atomic values and no repeating groups. 2NF: All non-key attributes depend on the whole key. 3NF: No transitive dependencies; user, post, and tag data are stored only in their respective tables.